

# **CURRICULUM VITAE**

## **PERSONAL PROFILE**

Given Name : NITHYANANDAN  
Family Name : KANAGARAJ  
Mothers Name : PARVATHY  
Date of Birth : 04.05.1986  
Nationality : Indian  
Gender : Male  
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Scopus Author ID : [36246432800](https://www.scopus.com/authid/detail/authid/36246432800)  
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\*All IDs are hyperlinked to the web urls for easy access

## 1. EDUCATIONAL QUALIFICATIONS

| <i>S. No</i> | <i>Degree</i> | <i>Subject</i>                              | <i>Class &amp; Percentage</i>   | <i>Year</i> | <i>Board/University</i>   |
|--------------|---------------|---------------------------------------------|---------------------------------|-------------|---------------------------|
| 1.           | X             | Mathematics, Science and social science     | I Class with distinction<br>92% | 2001        | State board of Tamil Nadu |
| 2.           | XII           | Mathematics, Physics, Chemistry and Biology | I Class<br>83%                  | 2003        | State board of Tamil Nadu |
| 3.           | B. Sc.        | Physics (Main)<br>Mathematics<br>Chemistry  | I Class<br>76%                  | 2006        | Pondicherry University    |
| 4.           | M. Sc.        | Physics (Material Science)                  | I Class<br>8.1 CGPA             | 2008        | Pondicherry University    |

## 2. Research Experience:

| <i>S. No.</i> | <i>Designation</i>       | <i>University/Institute</i>                                    | <i>Duration</i> |            |
|---------------|--------------------------|----------------------------------------------------------------|-----------------|------------|
| 1             | Visiting Researcher      | Institute of Photonics and Electronics, Prague, Czech Republic | 17-10-2025      | 25-10-2025 |
| 2             | Visiting Researcher      | Universite de Bourgogne, France                                | 18-06-2025      | 25-07-2025 |
| 3             | Visiting Researcher      | University of Southampton                                      | 13-03-2025      | 24-03-2025 |
| 4             | Research Fellow          | Optoelectronic Research Centre, Southampton                    | 09-09-2019      | 31-12-2020 |
| 5             | Researcher               | Institute of Photonics and Electronics, Prague                 | 01-01-2019      | 31-08-2019 |
| 6             | CNRS Postdoctoral Fellow | LIPHY Universite de Grenoble-Alpes, Grenoble, France           | 01-11-2017      | 31-12-2018 |
| 7             | CNRS Postdoctoral Fellow | Universite de Bourgogne, Dijon, France                         | 27-05-2016      | 31-10-2017 |
| 8             | CSIR SRF Extended (RA)   | Pondicherry University, Puducherry                             | 01-04-2015      | 01-04-2016 |
| 9             | CSIR SRF                 | Pondicherry University, Puducherry                             | 01-04-2013      | 01-04-2015 |

|    |                  |                                    |            |            |
|----|------------------|------------------------------------|------------|------------|
| 10 | Research Scholar | Pondicherry University, Puducherry | 27-08-2010 | 27/04/2015 |
| 11 | Project Fellow   | Pondicherry University, Puducherry | 04-05-2010 | 31-08-2010 |
| 12 | Project Fellow   | Pondicherry University, Puducherry | 28-07-2008 | 15-06-2009 |

### 3. Research Statistics

| International Journal Publications |    |
|------------------------------------|----|
| Published                          | 60 |
| In Review                          | 6  |
| To be submitted                    | 2  |

| Conference Publications |    |
|-------------------------|----|
| National                | 30 |
| International           | 82 |

|                                         |
|-----------------------------------------|
| <b>Overall Research Summary</b>         |
| 18 years                                |
| <b>Areas of Interest:</b>               |
| Ultrafast Optics                        |
| Lasers and Photonics                    |
| Fiber Optics Communications             |
| Laser for defence, space and industries |
| <b>No. of Publications:</b>             |
| In Journals : 60                        |
| In Conferences : 120                    |
| <b>h-index : 21 , I-index : 37</b>      |
| <b>Total Citations : 1892</b>           |

### 4. LIST OF PUBLICATIONS

#### A. List of International Journals

1. R. Vasantha Jayakantha Raja, K. Porsezian, and **K. Nithyanandan** “Modulational-instability induced supercontinuum generation with saturable nonlinear response”, ([Phys. Review A 82, 013825 \(2010\)](#)).
2. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, K. Porsezian, and B. Kalithasan, “Modulational instability with higher-order dispersion and walk-off effect in Kerr media with cross-phase modulation.”, ([Physical Review A 86, 023827 \(2012\)](#)).
3. K. Porsezian, **K. Nithyanandan**, R. Vasantha Jayakantha Raja, and P. K. Shukla, “Modulational instability at the proximity of zero dispersion wavelength in the relaxing saturable nonlinear system” ([Journal of Optical society of America B, 29, 2803 \(2012\)](#)).
4. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, K. Porsezian, “A theoretical investigation of Modulational instability in the novel semiconductor doped dispersion decreasing fiber and its cutting-edge over the existing fiber systems”, [Journal of Optical society of America B, 30, 178 \(2013\)](#).
5. **K. Nithyanandan**, and K. Porsezian “Interplay between relaxation of nonlinear response and coupling coefficient dispersion in the instability spectra of dual core optical fiber”, [Optics Communications, 303, 46 \(2013\)](#).

6. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, and K. Porsezian “*A colloquium on the influence of versatile class of saturable nonlinear responses on the white light lasers*”. [Optical Fiber Technology 19, 348 \(2013\).](#)
7. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, and K. Porsezian “*Modulational Instability in a twin core fiber with the effect of saturable nonlinear response and coupling coefficient dispersion*” ([Physical Review A 87, 043805 \(2013\).](#)
8. K. Porsezian, **K. Nithyanandan**, R. Vasantha Jayakantha Raja, and R. Ganapathy, “*A theoretical investigation of soliton induced supercontinuum generation in liquid core photonic crystal fiber and dual core optical fiber*” [European Physics Journal –Special Topics 222, 625 \(2013\).](#)
9. **K. Nithyanandan**, and K. Porsezian “*Observation of two state behavior in the instability spectrum of saturable nonlinear media*”, [European Physics Journal –Special Topics 222, 821 \(2013\).](#)
10. T. Uthayakumar, R. Vasantha Jayakantha Raja, **K. Nithyanandan**, and K. Porsezian, “*Designing a class of asymmetric twin core photonic crystal fibers for switching and multi-frequency generation*”, [Optical Fiber Technology 19 556 \(2013\).](#)
11. **K. Nithyanandan** and K. Porsezian, “*Impact of higher order dispersion on Modulational instability spectrum of a relaxing coupled saturable media*”, [Pramana–J. Phys 82 \(2\), 339 \(2014\).](#)
12. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, and K. Porsezian, “*Power play in the supercontinuum spectra of saturable nonlinear media*”. [Laser Physics 24, 045405 \(2014\).](#)
13. A. K. Shafeeque Ali, **K. Nithyanandan** and K. Porsezian, “*A Theoretical Investigation of Modulational Instability in three core Coupler with Negative Index Material channel*” [Physics Letters A 379\(3\) 223 \(2015\).](#)
14. **K. Nithyanandan** and K. Porsezian, “*A Novel Behavior of Pump Power in the Instability Induced Supercontinuum Generation of Saturable Nonlinear Media*”, ([J. of Physics, 605 \(2015\) 012031\).](#)
15. **K. Nithyanandan** and K. Porsezian, “*Influence of the functional form of nonlinearity in the Modulational Instability spectra of relaxing saturable nonlinear system*”, ([J. of Physics, 605 \(2015\) 012032\).](#)
16. **K. Nithyanandan** and K. Porsezian “*Co-propagation of two optical fields in the semiconductor doped dispersion decreasing fiber and modulational instability induced by cross-phase modulation*”. ([Laser Physics 1, 015401, \(2016\).](#)
17. A. K. Shafeeque Ali, **K. Nithyanandan** and K. Porsezian “*Modulational Instability in Triangular three-core Coupler with Negative Index Material channel*” [Journal of Optics 18 \(2016\) 035102\).](#)
18. A. K. Shafeeque Ali, **K. Nithyanandan** and K. Porsezian “*Influence of Birefringence in the Instability Spectra of Oppositely Directed Coupler with Negative Index Material Channel*”. [Physical Review A, 93, 023848 \(2016\).](#)

19. A. K. Shafeeque Ali, **K. Nithyanandan** and K. Porsezian “*A Theoretical Study on Threshold Conditions of Modulation Instability in Oppositely Directed Coupler*”. [Journal of Optics 18 \(12\) 125502, 2016](#).
20. A. K. Shafeeque Ali, **K. Nithyanandan** and K. Porsezian and PT Dinda “*Switching and Optical Bistability in Oppositely directed coupler*”. [IEEE J. of Quantum Electronics 52 \(12\), 1-8, 2016](#).
21. N.M Musamil, K. Porsezian, P.A. Subha, and **K. Nithyanandan** “*Dynamics of dark solitons propagation and tunneling through the variable coefficient coupled nonlinear Schrodinger equation*” [Chaos 27\(2\) 023113, 2017](#).
22. N,M Musamil, K. Porsezian, **K. Nithyanandan** and P.A. Subha, “*Dynamics of ultra-short dark solitons in variable coefficient modified nonlinear Schrödinger equation in nonlinear Optics*” [Optical fiber technology 37 11-20 \(2017\)](#).
23. S. Bhuvaneswari, **K. Nithyanandan**, P. Muruganandham, and K. Porsezian, “*Modulation instability in quasi two-dimensional spin-orbit coupled Bose-Einstein condensates*” ([J. Phys. B: At. Mol. Opt. Physics 49 \(24\), 245301, 2016](#)).
24. Katryna Krupa **K. Nithyanandan** Ugo Andral, P.T Dinda, Philips Grelu, “*Direct experimental evidence of soliton molecular motions*” [Physical Review Letters 118, 243901 \(2017\)](#).
25. **K. Nithyanandan** A. K. Shafeeque Ali, K. Porsezian, P.T. Dinda and P. Grelu, “*A novel observation of Multistability in oppositely directed coupler*, [Optics communications 416, 145, 2018](#)).
26. Katryna Krupa **K. Nithyanandan**, Philippe Grelu, “*Vector dynamics of incoherent dissipative solitons*” ([Optica 10, 1239, 2017](#)).
27. N,M Musamil, P.A. Subha and **K. Nithyanandan**, “*Black and gray soliton interactions and cascade compression in the variable coefficient nonlinear Schrodinger equation*” [Optik 159, 176, 2018](#)).
28. Monisha Kumar, **K. Nithyanandan**, and K.Porsezian “*Influence of spatial delay on the modulational instability in a composite system with a controllable nonlinearity*, [Physical Review E, 97 062208 \(2018\)](#)).
29. S. Bhuvaneswari, **K. Nithyanandan**, and P. Muruganandam, “*Spotlighting phase separation and tricriticality in Rasbha spin-orbit coupled Bose-Einstein condensates in two dimensions*” ([Journal of Physics Communications 2 025008, 2018](#)).
30. **S. Sharaf Ali**, **K. Nithyanandan** and K. Porsezian “*Self-similar pulse compression by defective core photonic crystal fiber with cubic-quintic nonlinearities*”, ([Optik, 178, 591-601, 2018](#)).
31. Zhiqiang Wang, **K. Nithyanandan**, A. Coillet , Philippe Grelu, “*Real-time measurements of internal dynamics of bound soliton molecules in a passively mode-locked fiber laser*” ([Nature Communications 10\(1\),830 2019](#) ).
32. Monisha Kumar, **K. Nithyanandan**, and K. Porsezian “*Instabilities and solitons in systems with spatio temporal dispersions and non paraxial approximations*” ([Optik, 182 1120 \(2019\)](#)).

33. N,M Musamil, P.A. Subha and **K. Nithyanandan**, “Phase dynamics of inhomogeneous Manakov vector solitons” [\*Physical Review E\* 100, 012213, \(2019\).](#)
34. J. Igbancho, **K.Nithyanandan**, P.T. Dinda, P. Grelu “**Dynamics of distorted and undistorted soliton molecules in a mode-locked fiber laser**” ([\*Physical Review A\* 99, 063824 \(2019\).](#)
35. **K. Nithyanandan**, L. Djevarhidjian, V. Duran Bosch, C. Schnébelin, S. Kassi, G. Méjean, D. Romanini, and H. Guillet de Chatellus “*Optimization of acousto-optic frequency combs for multi heterodyne spectroscopy* ([\*Optics Express\* 27 14842-14852 \(2019\).](#)
36. **K. Nithyanandan**, Antreas Theodosiou, Jan Aubrecht, Pavel Peterka, Kyriacos Kalli, I. Kašik and Pavel Honzátko, *All-Fiber Mode-Locked Thulium Doped Fiber Laser Using a Novel Femtosecond Laser Inscribed 450 Tilted Fiber Grating* ([\*Laser Physics Letters\* – 16, 0951904 \(2019\)](#)
37. A. B. Moubissi, Th. B. Ekogo, Moussambi, Membetsi, **K. Nithyanandan** and S. Nse Biyoghe, “Energy diagram and stability range of families of SMs in fibers” [\*Optical and Quantum Electronics\* 51 \(5\), 160 \(2019\).](#)
38. S. Sharaf Ali, **K. Nithyanandan** “*Ultrabroad continuum generation in silic based defective core photonic crystal fiber*” ([\*Optik\*, 191 \(2019\) 121 – 131\).](#)
39. H. Triki, K. Porsezian, K. Senthilnathan, and **K.Nithyanandan**, *Chirped self-similar solitary waves for the generalized nonlinear Schrödinger equation with distributed two power-law nonlinearities* [\*Physical review E\* 100 042208 \(2019\).](#)
40. V. Duran Bosch, H. Guillet de Chatellus, L. Djevarhidjian, **K. Nithyanandan**, , C. Schnébelin, S. Kassi, G. Méjean, D. Romanini “*Optical frequency combs generated by acoustic optic frequency shifting loops*” ([\*IEEE Photonics Technology Letters\*, 13, 1878 \(2019\).](#)
41. Zhiqiang Wang, **K. Nithyanandan**, Philippe Grelu, “*Build-up of Noise like pulse emission in fiber laser cavity*, ([\*Physical Review Research\*, 2 013101, \(2020\)\).](#)
42. **S. Sharaf Ali, K. Nithyanandan** “A theoretical study on the Supercontinuum generation in a novel suspended liquid core photonic crystal fiber” ([\*Applied Physics B\* 126, 55 \(2020\).](#)
43. Vincent Billault, Vincent Crozatier, Ghaya Baili, Loic Morvan, Daniel Dolfi, Fellow, OSA, **K.Nithyanadan** , Fellow, OSA, and Hugues Guillet de Chatellus, “Phase Noise of Optical Pulse Trains Generated by Talbot Effect in Frequency Shifting Loops”, ([\*J. of Lightwave Techn.\* 39 \(8\) 2336-2347 \(2021\).](#)
44. Filip Todorov, Jan Aubrecht, ,Pavel Peterka, Ondřej Schreiber, **K. Nithyanandan**, P. Peterka, “Active Optical Fibers and Components for Fiber Lasers Emitting in the 2-μm Spectral Range, [\*Materials\*, 2020 13\(22\) 5177.](#)
45. M. Grabner, **K. Nithyanandan**, P. Peterka, P. Koska, A. A. Jasim, and P. Honzatko, “Simulations of Pump Absorption in Tandem-Pumped Octagon Double-Clad Fiber Lasers, ([\*IEEE Photonics letters\*, 13 \(2\) 1-14 \(2021\).](#)
46. Houria Triki, Amala Jose and K. Nithyanandan, “*Chirped Self-similar localized pulses on a continuous wave background in presence of cubic-quintic nonlinearity and self-frequency shift*” [\*Optik\* 270, 169876, 2022.](#)

47. Z Wang, Q Jiang, N Kanagaraj, B Yu, Z Zhang, “Spotlighting the Simultaneous Formation of Coherent and Incoherent Dissipative Solitons in an Er-Doped Bidirectional Ultrafast Fiber Laser” [Physical Review Applied 18 \(6\), 064096, 2022.](#)
48. N Tcheppen, S Balasubramanian, K. **Nithyanandan** “Modulational instability in a coupled nonlocal media with cubic, quintic and septimal nonlinearities, [Nonlinear Dynamics 111 \(21\), 20311-20329, 2023.](#)
49. A Jose, **K Nithyanandan**, “A Comparative Study on the Effect of dispersion in Noise-Like Pulse Pumped Supercontinuum Generation, [Springer Nature 219-222, 2024.](#)
50. **K. Nithyanandan**, “Soliton Physics in India: A tribute to the late K. Porsezian, [Optics Communications 553, 130078, 2024.](#)
51. Krushna Devkar; Arunkumar S; Tithi Bhakta; Shrikant R. Bharadwaj; Nithyanandan. K; Nagarajan Ganapathy, “Explainable XceptionNet-Based Keratoconus Detection Using Infrared Images and Edge Computing” [IEEE Sensor Letters, 2475-1472, 2024, \(Early Access\) I.F. 2651.](#)
52. Amala Jose, SD Chowdhury, S Balasubramanian, K Krupa, Z wang, P Grelu, K. Nithyanandan, “Noise-Like Pulse Seeded Supercontinuum Generation: An In-Depth Review For High-Energy Flat Broadband Sources, [Laser & Photonics Review, 2400511, 2025](#)
53. T Akash Domnic, S Khushboo, K Nithyanandan, “Physics-Based Phase control in Coherent Beam Combining Systems”, [Journal of Lightwave Technology, 43 \(2\) 824-831, 2025.](#)
54. Amala Jose, M. Guasoni, A. Coillet, PT Dinda, P Grelu, K Nithyanandan, “Towards a broadband gain-flattened thulium doped fiber amplifier”, [Optics Commun. 578, 131452, 2025.](#)
55. Jack Haines, Pooja Naik, K. Nithyanandan, Massimiliano Guasoni, “Integrated  $1 \times N$  Polarization-Insensitive Optical Power Splitters,” [Optics Letters 50 \(17\), 5278-5281, 2025.](#)
56. M. Dinesh Kumar, Rajkumar Joseph, K. Nithyanandan, “Transformer Encoder–Decoder Framework for Nonlinear Pulse Evolution and Inverse Modeling”, [Advanced Photonics Research, 2500149, 2025](#)
57. Mostafa Mohammad, Amala Jose, Aurelien Coillet, PT Dinda, Nithyanandan Kanagaraj, Philippe Grelu, “Highly dispersive energy managed soliton fiber laser” , [Optics Letters 50 \(19\), 6101-6104, 2025.](#)
58. Bharathy Jacob, Nithyanandan Kanagaraj, Yogesh Natesan, Pooja Uday Naik, Aneesh Vincent Veluthandath, Massimiliano Guasoni, “Polarization-Dependent Cylindrical Vector Vortex Beams and Optical Skyrmion Generation Using All-Dielectric Chiral Metasurfaces” , [Physical Review A \(Article in Press\)](#)
59. Mukesh Kumar and K. Nithyanandan, “Effects on beam quality due to misalignment errors in the beam combination systems,” Optics and Laser Technology ([Accepted with Minor corrections](#)).
60. Divyadarshini Madhavan, Krushna Devkar, Himanshu Singh, Nithyanandan Kanagaraj, Shrikant R. Bharadwaj, Nagarajan Ganapathi, “KERALite: An end-to-end Keratoconus



*Detection using Ultra-lightweight 2D Convolution Neural Networks,” Biomedical Signal Processing and Control, BSPC-D-25-11015, 2025.*

61. Khushboo Soni, S Thirumugam, John Rozario Jegaraj, Nithyanandan Kanagaraj, “*Adaptive Laser Beam Engineering with Coherent Beam Combining for Efficient and Uniform Power Delivery*” , [Journal of Lightwave Technology \(Under Review\)](#). (arXiv preprint arXiv:2510.19464).
62. Bharathy Jacob, John Rozario Jegaraj, Nithyanandan Kanagaraj, “*An all-Optical Passive Device based on Wavelength-Adaptive Diffractive Neural Network for Dynamic Beam Shaping*” [Optics and Laser Engineering \(under Review\)](#).
63. Subrata Manna and Nithyanandan Kanagaraj, “*Generation and Dynamics of Dark-Bright Optical Solitons in NPR-Based Mode-Locked Fiber Lasers: Formation of Dark-Bright-Dark Pulses and Multi-Soliton String*” [Optical Fiber Technology \(Under Review\)](#)
64. Subrata Manna and Nithyanandan Kanagaraj, “*Engineering Multiwavelength Emission in Mode-Locked Fiber Lasers via Nonlinear Polarization Rotation,*” [Optics and Laser Technology \(Under Review\)](#).
65. Mamatha K L, Stalin S, Nithyanandan Kanagaraj, “Propagation of Gaussian beam in strongly nonlocal nonlinear media with inhomogeneous diffraction” [Physics Letters A \(Under Review\)](#)
66. Dinesh Murugan and Nithyanandan Kanagaraj, “An approach to resolve modes in multimode fibre optics with transformer neural network,” [Advanced Photonics \(To be submitted\)](#).
67. Bharathy J, Akash Domnic Thomas, K. Nithyanandan “*Multitasking Optical Systems: Wavelength Demultiplexing Diffractive Neural Networks for Light Beam Shaping,*” [Optics and Laser Engineering \(under Review\)](#).
68. N. Nathan, Mamatha K L, and K. Nithyanandan, “*Optical soliton management with fourth-order dispersion and power-law nonlinearity in generalized P T -symmetric potential*”, (Communicated to Nonlinear Dynamics)
69. Bharathy Jacob ,John Rozario Jegaraj, Nithyanandan Kanagaraj, “*An all-Optical Passive Device based on Wavelength-Adaptive Diffractive Neural Network for Dynamic Beam Shaping*” [Optics and Laser Engineering \(under Review\)](#).
70. K. Nithyanandan, Katryna Krupa Philippe Grelu and J. Soto-Crespo, “*Exploration of vector dissipative soliton in the regime of noisy pulsation*”, **(In Preparation)**

## **B. List of papers in International Symposia /Conferences**

1. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian “Modulational Instability in the Regime of Minimum Group Velocity Dispersion for a Relaxational Saturable Medium” ICONTOP proceedings (ISBN 978-81-908188-1-0), 290-295 (2011) , in International Conference on Trends in Optics and Photonics held on Dec. 7-9 (2011) in University of Calcutta.



2. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian “*Modulational Instability in the Regime of Minimum Group Velocity Dispersion for a Relaxational Saturable Medium*” ICONTOP proceedings (ISBN 978-81-908188-1-0), 290-295 (2011) , in International Conference on Trends in Optics and Photonics held on Dec. 7-9 (2011) in University of Calcutta.
3. T. Uthayakumar, R. Vasantha Jayakantha Raja, **K. Nithyanandan**, and K. Porsezian, “*On the Propagation of Light through a Twin Core Photonic Crystal Fiber at 850 nm*” ICONTOP proceedings (ISBN 978-81-908188-1-0), 370-375 (2011), in International Conference on Trends in Optics and Photonics held on December 7-9 (2011) in University of Calcutta.
4. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian “*Interplay between saturation and relaxation of nonlinear response in the Modulational Instability of various nonlinear media*”, in the International conference International conference on Optical Engineering (ICOE2012) held on July 26-28 (2012) at Visvesvaraya Technological University (VTU), Belgaum, India. (DOI: 10.1109/ICOE.2012.6409578).
5. T. Uthayakumar, R. Vasantha Jayakantha Raja, **K. Nithyanandan**, and K. Porsezian, “*Investigation of optical pulse coupling through the nitrobenzene filled photonic crystal fiber coupler-A projection operator approach*”, in the International conference International conference on Optical Engineering (ICOE2012) held on July 26-28 (2012) at Visvesvaraya Technological University (VTU), Belgaum, India. (DOI: 10.1109/ICOE.2012.6409580)
6. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*Modulation Instability in Optical Fibers in Relaxing Kerr medium*” in the second International Optics Network of Students (IONS) conference held on December 07-08, 2012 at IIT Madras, Chennai, India.
7. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*A Unique Behavior of Pump power in the Modulational Instability Spectra of Saturable Nonlinear Media*”, in the Photonics 2012, International Conference on Fiber Optics and Photonics, held on December 09-12, 2012 at IIT Madras, Chennai, India. (DOI: 10.1364/PHOTONICS.2012.Wpo.4)
8. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*Impact of Dispersion Profile on Modulational Instability in Dispersion Decreasing Fiber*”, in the Photonics 2012, International Conference on Fiber Optics and Photonics, held on December 09-12, 2012 at IIT Madras, Chennai, India. (DOI : 10.1364/PHOTONICS.2012.Wpo.1)
9. Vasantha Jayakantha Raja, **K. Nithyanandan**, T. Uthayakumar, and K. Porsezian, “*Influence of Material Absorption on Supercontinuum Generation in Photonic Crystal Fibers*”, in the Photonics 2012, International Conference on Fiber Optics and Photonics, held on December 09-12, 2012 at IIT Madras, Chennai, India. (DOI: 10.1364/PHOTONICS.2012.Wpo.2)
10. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*Impact of Material Absorption on Supercontinuum generation in liquid core photonic crystal fiber*” published in the AIP proceeding by the International Conference on recent trends in applied

Physics (RAM 2013) , held on Feb 1 & 2, 2013 at Govt. College of Engineering and Technology, Bikaner, Rajasthan, India. (DOI: 10.1063/1.4810426)

11. T. Uthayakumar, R. Vasantha Jayakantha Raja, **K. Nithyanandan**, and K. Porsezian, “*All Optical AND And NAND Logic Gates Based On A Triple Core Photonic Crystal Fiber*” published in the AIP proceeding by the International Conference on recent trends in applied Physics (RAM 2013) , held on Feb 1 & 2, 2013 at Govt. College of Engineering and Technology, Bikaner, Rajasthan, India. (DOI: 10.1063/1.4810427).
12. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*A novel power play in the supercontinuum generation of saturable nonlinear optical media*”, published in the abstract – page no. 68 in the International conference on “Perspectivies on Nonlinear Dynamics” (PNLD – 2013) held on July 15 – 18, 2013 at Hyderabad Central University (HCU), Hyderabad.
13. T. Uthayakumar, R. Vasantha Jayakantha Raja, **K. Nithyanandan**, and K. Porsezian, “*Investigation of Transmission characteristics of Triple Core Microstructured fiber for optical logic gates,*” published in the abstract – page no. 86 in the International conference on “Perspectivies on Nonlinear Dynamics” (PNLD – 2013) held on July 15 – 18, 2013 at Hyderabad Central University (HCU), Hyderabad, India.
14. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*Influence of various functional form of saturable nonlinearity in the instability spectra of delayed nonlinear system*” in the 8<sup>th</sup> International Conference of Nonlinear System and Dynamics (CNSD – 2013) held on December 11 – 14, 2013 at Indian Institute of Technology (IIT) Indore, Indore, India.
15. T. Uthayakumar, R. Vasantha Jayakantha Raja, **K. Nithyanandan**, and K. Porsezian,, “*Investigation of the role of asymmetry in all optical steering behavior of photonic crystal fiber coupler*” in the 8<sup>th</sup> International Conference of Nonlinear System and Dynamics (CNSD – 2013) held on December 11 – 14, 2013 at Indian Institute of Technology (IIT) Indore, Indore, India.
16. **K. Nithyanandan**, and K. Porsezian, “*Supercontinuum generation in saturable nonlinear media*” in the IONS -7 North America held on Oct 2 – 4, 2013 at University of North Carolina, Charlotte, United States of America. USA.
17. **K. Nithyanandan**, and K. Porsezian, “*A colloquium on the Ultrashort Nonlinear pulse propagation through Optical fibers: Modulational Instability and Supercontinuum generation*”, in the KOALA Conference held on November 24 – 29, 2013 at University of Sydney, Sydney, Australia.
18. **K. Nithyanandan**, and K. Porsezian, “*Observation of two state behavior in the Instability induced Supercontinuum Generation of Exponential Saturable Nonlinearity*” in the Workshop on Recent Advances in Physics (WRAP – 2013) held on December 17 – 18, 2013 at Indian Institute of Technology (IIT) Delhi, Delhi, India. (DOI: 10.1109/WRAP.2013.6917695)

19. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*Modulation Instability in Optical fibers in relaxing nonlinear media*” (PP-NQQ-10) published in the proceedings of International conference on Optics and Opto-electronics (ICOL -2014), held on March 05– 08, 2014 at Instrument Research and Development Establishment (IRDE), Dehradun, India.
20. T. Uthayakumar, R. Vasantha Jayakantha Raja, **K. Nithyanandan**, and K. Porsezian, “*CS2 Filled Triangular Triple core PCF for All Optical Logic circuits*”, (PP-FIO-14) published in the proceedings of in the International conference on Optics and Opto-electronics (ICOL -2014), held on March 05– 08, 2014 at Instrument Research and Development Establishment (IRDE), Dehradun.
21. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*A theoretical investigation on super continuum generation in exponential saturable nonlinear response*”, (PP-NQQ-9) published in the proceedings of in the International conference on Optics and Opto-electronics (ICOL -2014), held on March 05– 08, 2014 at Instrument Research and Development Establishment (IRDE), Dehradun, India.
22. **K. Nithyanandan** and K. Porsezian, “*A Novel Behavior of Pump Power in the Instability Induced Supercontinuum Generation of Saturable Nonlinear Media*”, in the 23<sup>rd</sup> Congress of International Commission for Optics (ICO -23) held on 26 – 29, August, 2014 at Santiago de Compostela, Spain (**Journal of Physics: Conferences Series**).
23. **K. Nithyanandan** and K. Porsezian, “*Influence of the functional form of nonlinearity in the Modulational Instability spectra of relaxing saturable nonlinear system*”, in the 23<sup>rd</sup> Congress of International Commission for Optics (ICO -23) held on 26 – 29, August, 2014 at Santiago de Compostela, Spain. (**Journal of Physics: Conferences Series**).
24. **K. Nithyanandan** and K. Porsezian, “*A Theoretical Investigation on Modulational Instability in non-instantaneous Saturable Nonlinear Media*”, Published in the technical digest (17p\_C4\_8) in the JSAP-OSA Joint Symposia (JSAP) 2014 held on 17 – 20 September 2014 at Sapporo Campus, Hokkaido University, Hokkaido, Japan. (**DOI: 10.1364/JSAP.2014.17p\_C4\_8**).
25. **K. Nithyanandan** and K. Porsezian, “*A Study on Super Continuum Generation in Exponential type Saturable Nonlinearity*”, Published in the technical digest (17p\_C4\_9) in the JSAP-OSA Joint Symposia (JSAP) 2014 held on 17 – 20 September 2014 at Sapporo Campus, Hokkaido University, Hokkaido, Japan. (**DOI: 10.1364/JSAP.2014.17p\_C4\_9**).
26. **K. Nithyanandan** and K. Porsezian, “*A Novel Power play in the Supercontinuum Generation Induced by Modulational Instability in Saturable Nonlinear Media*”, Published in the technical digest (FW4D.7 – ISBN 1557522863) in the Frontiers in Optics/Laser Science (FIO-2014) held on 19 – 23, October, 2014 at Tucson, Arizona, United States of America.
27. **K. Nithyanandan**, and K. Porsezian, “*A colloquium on the Ultrashort Nonlinear pulse propagation through Optical fibers: Modulational Instability and Supercontinuum*

generation”, in the KOALA Conference held on November 23 – 28, 2014 at University of Adelaide, Adelaide, Australia.

28. **K. Nithyanandan**, A. K. Shafeeque Ali and K. Porsezian, “*Modulational instability in the semiconductor doped dispersion decreasing fiber – A prospect for ultrashort pulse generation*”, 12<sup>th</sup> International Conference on Fiber Optics and Photonics, held on December 16, 2014 at IIT Kharagpur, Kharagpur, India. (DOI: **10.1364/PHOTONICS.2014.T3A.25**).
29. **K. Nithyanandan**, “*A Theoretical observation of novel power play in the Supercontinuum spectra of exponential Saturable nonlinear media*”, in the topical meeting on Nonlinear Optics 2015, held on, 26-31 July 2015 at Kauai, Hawaii, United States, (DOI: **10.1364/NLO.2015.NW4A.1**) .
30. **K. Nithyanandan**, A. K. Shafeeque Ali and K. Porsezian, “*A Unique Observation of Power Play in the Instability Induced Supercontinuum Generation in Saturable Nonlinear Media*”, in the Frontiers in Optics/Laser Science (FIO-2016) held on 18 – 22, Oct, 2015 at San Jose, United States of America.
31. **K. Nithyanandan**, “*A Novel Power play in the Supercontinuum generation induced by Modulational instability in Saturable nonlinear Media*”, in the Advanced Photonics 2016, Specialty optical fibers 2016, Vancouver Canada, 18-20 July 2016. (DOI: **10.1364/SOF.2016.SoW3G.6**) .
32. **K. Nithyanandan**, A. K. Shafeeque Ali and K. Porsezian, “*A Novel Observation of Optical Bistability in Oppositely Directed Coupler*”, in the Frontiers in Optics/Laser Science (FIO-2016) JW4A. 169 held on 17 – 21, October, 2016 at Rochester, New York, United States of America. (DOI: **10.1364/FIO.2016.JW4A.169**).
33. **K. Nithyanandan**, A. K. Shafeeque Ali and K. Porsezian, “*Observation of two state behavior in the Supercontinuum Generation induced by instability in Saturable Nonlinear Media*”, in the Laser Science (FIO-2016) JW4A. 46 held on 17 – 21, October, 2016 at Rochester, New York, United States of America (DOI: **10.1364/FIO.2016.JW4A.46**)
34. **K. Nithyanandan** and K. Porsezian, “*A study on the Modulational Instability in delayed Saturable Nonlinear Optical Media*”, Published in the technical digest (16p\_C301\_11) in the JSAP-OSA Joint Symposia held on 13 – 16 September 2016 at Niigata Japan, Japan. (ISBN: **978-4-86348-589-1**).
35. **K. Nithyanandan** and K. Porsezian, “*A Theoretical Investigation on Supercontinuum Generation in Saturable Optical Nonlinear Media*”, Published in the technical digest (16a\_C301\_9) in the JSAP-OSA Joint Symposia held on 13 – 16 Sep. 2016 at Niigata Japan,. (ISBN: **978-4-86348-589-1**).
36. **K. Nithyanandan**, “*Optical Bistability in dual core directional coupler with negative index material channel*”, in the Laser Science (FIO-2017) JW4A. 123 held on 18 – 21, September, 2017 at Washington D. C United States of America (ISBN: **978-1-943580-33-0**)

37. **K. Nithyanandan**, “*Observation of Optical bistability in oppositely directed coupler*” Published in the technical digest (8a\_PB2\_7) in the JSAP-OSA Joint Symposia (JSAP) 2017 held on 18 – 21 September 2017 at Fukuoka Japan. (ISBN: 9784863486416).
38. **K. Nithyanandan**, R. Vasantha Jayakantha Raja, T. Uthayakumar, and K. Porsezian, “*Modulational Instability in a composite system with nonlinearity showing rectangular response*” (PP-NQQ-10) published in the proceedings of International conference on Optics and Opto-electronics (ICOL -2014), held on March 05– 08, 2014 at Instrument Research and Development Establishment (IRDE), Dehradun, India.
39. K. Krupa, **K. Nithyanandan**, Ugo Andral, P.T. Dinda and Ph. Grelu, “*Observation temps réel des oscillations ultrarapides de molécules de solitons dans un laser à fibre à modes bloqués*” published in the proceedings of JNOG 2017 held on 4 July 2017 at Limoges, France.
40. K. Krupa, **K. Nithyanandan**, and Ph. Grelu, “*Dynamique de solitons dissipatifs incohérents vectoriels*” published in the proceedings of JNOG 2017 held on 4 July 2017 at Limoges, France.
41. Monisha Kumar, **K. Nithyanandan**, and K. Porsezian, “*Modulation Instability in Optical fibers in relaxing nonlinear media*” (OP12) published in the proceedings of International conference on Advances in Optics and Photonics (ICOAP -2017), held on Nov 23– 26, 2017 at Guru Jambheshwar University of Science and Technology, Hisar, Harayana, India.
42. S.Sharaf, K. Porsezian, and **K. Nithyanandan**, “*Self-Similar Pulse Compression and Supercontinuum Generation in CS<sub>2</sub> filled Photonic crystal fibers*” (PP59) published in the proceedings of International conference on Advances in Optics and Photonics (ICOAP -2017), held on Nov 23– 26, 2017 at Guru Jambheshwar University of Science and Technology, Hisar, Harayana.
43. Monisha Kumar, K. Porsezian and K. Nithyanandan, “*Dispersion relations and band gaps in wave number or frequency in the linear and nonlinear regimes for a coupled system with no paraxial approximation*”, AIP conference Proceedings 1942(1) 110053.
44. V. Duran Bosch, L. Djevarhidjian, C. Schnébelin, K. Nithyanandan, S. Kassi, G. Méjean, D. Romanini, and H. Guillet de Chatellus, “*Self-heterodyne and dual-comb spectroscopy using acousto-optic frequency combs*”, published in CLEO: Applications and Technology 2018, held on 13-18 May, held at San Jose, California, United States (DOI: 10.1364/CLEO\_AT.2018.ATh3O.4).
45. **K. Nithyanandan**, “*Bistability in Oppositely directed coupler with negative index material channel*”, in the Advanced photonics meeting (JTU5A.59) to be held on 02 – 05, July, 2018 at Zurich. (DOI: 10.1364/BGPPM.2018.JTu5A.59).
46. J. Igbanoacho, **K. Nithyanandan** K. Krupa. P.T. Dinda, Ph. Grelu, “*Vibrations and oscillations of Tri-soliton Molecules in a Mode-locked fiber laser,*” (NpW2C.5) to be held on 02 – 05, July, 2018 Zurich (DOI:10.1364/NP.2018.NpW2C.5).
47. **K. Nithyanandan**, L. Djevarhidjian, V. Duran Bosch, C. Schnébelin, S. Kassi, G. Méjean, D. Romanini, and H. Guillet de Chatellus “*Optimization of acousto-optic frequency combs for*

*multi-heterodyne spectroscopy*”, (LTh1F.3) published in *Frontiers in Optics/Laser Science (FIO -2018)*, held on 16-20 September 2018, held at Washington DC, United States (ISBN :978-1-943580-46-0).

48. **K. Nithyanandan**, “*Bistability in Oppositely directed coupler with negative index material channel*”, (JW4A. 86) held on 16-20 September 2018, held at Washington DC, United States (ISBN :978-1-943580-46-0).
49. Ph. Grelu , Z. Wang, S. Hamdi, **K. Nithyanandan**, Aurelien Coillet, P.T Dinda, “*New lights from ultrafast fiber laser real-time recordings*,” EuroPhoton 2018, held on 02-07 Sep. 2018 at Barcelona.
50. **K. Nithyanandan**, L. Djevarhidjian, V. Duran Bosch, C. Schnébelin, S. Kassi, G. Méjean, D. Romanini, and H. Guillet de Chatellus “*Optimization of acousto-optic frequency combs (CLEO Europe -2019)*”, held on 23-27 June 2019, held at Munich, Germany.
51. **K. Nithyanandan**, Z. Wang, Aurelien Coillet, P.T Dinda, Ph. Grelu “*Build-up of Noise-like Pulses in Ultrafast Fiber Lasers, (CLEO Europe -2019)*”, held on 23-27 June 2019, held at Munich, Germany.
52. Antreas Theodosiou, Jan Aubrecht, **K. Nithyanandan**, Pavel Peterka, Kyriacos Kalli, Ivan Kašík and Pavel Honzátko, “*All-fiber mode-locked thulim doped fiber laser using a novel femtosecond laser inscribed 450-Tilted fiber grating, (CLEO Europe -2019)*”, held on 23-27 June 2019, held at Munich, Germany.
53. **K. Nithyanandan**, Z. Wang, Aurelien Coillet, P.T Dinda, Ph. Grelu “*Optical soliton molecular complexes in ultrafast fiber lasers, (CLEO Europe -2019)*”, held on 23-27 June 2019, held at Munich, Germany.
54. Ph. Grelu , Z. Wang, S. Hamdi, **K. Nithyanandan**, Aurelien Coillet, P.T Dinda, “*Complex ultrafast fiber laser dynamics in real-time, Progress in Electromagnetic Research Symposium(PIERS -2019)*”, held on 17-20 June 2019, held at Rome, Italy.
55. A. Theodosiou, J. Aubrecht, **K. Nithyanandan**, K. Kalli, A. Stassis, P. Peterka, I. Kasik, P. Honzatko, “*Monolithic fibre lasers developed using the plane-by-plane femtosecond laser inscription method, International microwave and Optoelectronics conference*”, Nov, 12 -14, Aveiro, Portugal.
56. **K. Nithyanandan**, A. Sharafali, and P.Peterka “*A theoretical study on the continuum generation in a defective core photonic crystal fiber*”, OSA Laser congress, Sep 29 – Oct 4, Vienna, Austria.
57. M. Grábner, **K.Nithyanandan**, P. Peterka, P. Koška, P. Honzátko, and A. A. Jasim, “*Numerical modeling of pump absorption in coiled and twisted double-clad fiber: a prospect for tandem pumped fiber laser*,” in Proc. SPIE 11355, SPIE Photonics Europe, p. 113550W.
58. M. Grábner, **K. Nithyanandan**, P. Peterka , P. Koška, A. A. Jasim, P. Honzátko, “*Numerical study of effect of bending and twist on pump absorption in octagonal double-clad fiber*” Europhotonics, Prague.
59. Amala Jose and **K. Nithyanandan**, “*Theoretical investigation of supercontinuum generation in photonic crystal fiber*” during the National Science Day, Feb 28, 2022.



60. Mamatha K. L., Amala Jose, and **K. Nithyanandan**, "*A recurrent neural network predicting the complex spatio temporal dynamics in fiber optics*" during the National Science Day, Feb 28, 2022.
61. Amala Jose, **K. Nithyanandan**, "*Impact of cavity dispersion on the Noise-like pulse generated supercontinuum*," *Frontiers in Optics 2022* held on 17–20 October 2022, Rochester, New York, United States.
62. Amala Jose, **K. Nithyanandan**, "*Effect of Noise-like pulse from opposite cavity dispersion on Supercontinuum generation*" 17th International Conference on Advanced Science and Technology, held on 1–2 December 2022, Kumamoto University, Japan.
63. Mamatha K L, Subrata Manna, Amala Jose, **K. Nithyanandan**, "*A study on the Self-Similar Solitary Wave Propagation in inhomogeneous Optical Fiber System*", CNSD National Conference, held in December 2022, IISER Pune, India.
64. Khushboo and **K. Nithyanandan**, "*High-Power Scaling: Coherent Beam Combining of Fiber Lasers*" during the National science day at IIT Hyderabad, Feb 2023.
65. Bharathy J, Yogesh Natesan, **K. Nithyanandan**, "*All-Dielectric Metamaterial Fano Resonators at Terahertz Waveband*", National Conference, held on 5–7 June 2023, Indian Institute of Technology Hyderabad, India.
66. Amala Jose, **K. Nithyanandan**, "*Investigation of Incoherent Soliton Liquid in the NLP Regime*" CNLDS 2023, held on 5–7 June 2023, IIT Hyderabad, India.
67. Mamatha K L, Amala Jose, **K. Nithyanandan**, "*Exploring Soliton Explosion in the Vector Dissipative Soliton Dynamics*," CNLDS – IITH, held on 5–7 June 2023, IITH, Hyderabad, India.
68. Amala Jose, **K. Nithyanandan**, "*A Comparative Study on the Effect of Dispersion in Noise-Like Pulse Pumped Supercontinuum Generation*," *PHOTONICS 2023* held on 5–8 July 2023, IISc Bangalore.
69. Amala Jose, **K. Nithyanandan**, "*Observation of novel incoherent dissipative solitons in polarization-resolved vector ultrafast fiber laser*," 84th JSAP Autumn Meeting, held on 19-23 September 2023, Kumamoto, Japan.
70. Khushboo, Akash Dominic Thomas and **K. Nithyanandan**, "*Coherent Beam Combining of the fiber lasers using Deep learning optimizers*" published in the Proceedings of the International Conference "Complexity and Nonlinear Dynamics in STEM" (CNLDS 2023), on pg. 95.
71. Kartik Kumar Kalia, Akash Dominic Thomas and **K. Nithyanandan**, "*Model-Free Prediction of Spectral Evolution in Optic Fibers Using Machine Learning*" published in the pg. 92, Proceedings of International Conference "Complexity and Nonlinear Dynamics in STEM" (CNLDS 2023), IIT Hyderabad.
72. Raghav Juyal and **K. Nithyanandan**, *Demonstrating Hybrid Quantum-Classical Neural Network*, published in the Proceedings of International Conference "Complexity and Nonlinear Dynamics in STEM" (CNLDS 2023), in the pg. 131.
73. Nathan Tchepemen, Sudharsan J B and **K. Nithyanandan**, *Modulational instability in a*

*coupled nonlocal media with cubic, quintic and septimal nonlinearities*, published in the Proceedings of International Conference “Complexity and Nonlinear Dynamics in STEM” in the pg. 98.

74. Bharadwaj, S. R., Madhavan, D., Bhakta, **K. Nithyanandan**, & Ganapathy, N. (2024). *A machine learning model for keratoconus detection using eccentric infrared photorefractive profiles*. American Academy of Optometry (AAO), Indianapolis, USA
75. Madhavan, D., Bharadwaj, S. R., **K. Nithyanandan**, & Ganapathy, N. (2024). *Artificial intelligence-based analysis of eccentric photorefractive profiles for keratoconus detection*. ResMeet '24, India Vision Institute, Chennai, India.
76. Amala Jose, **K. Nithyanandan**, “*Thermal Effect of a Passive Filter in a Two-stage Tm-doped Fiber Amplifier*”, International Conference , held on 23–25 October 2024, CSIR-CSIO, Chandigarh, India.
77. Khushboo Soni and **K. Nithyanandan** “*Effect of the Atmospheric Turbulence in the Propagation of the Coherent Combined Beam and Phase Synchronization using Deep Learning Optimizers*” International Conference held on 23–25 October 2024, CSIR-CSIO, Chandigarh, India.
78. Mamatha K L and **K. Nithyanandan** “*Impact of Inhomogeneous Diffraction effects on Beam Dynamics in Strongly Nonlocal Nonlinear Optical Media*” International Conference (Photonics, OSI Conferences), held on 23–25 Oct 2024, CSIR-CSIO, Chandigarh, India.
79. Bharathy J, Yogesh Natesan, **K. Nithyanandan**, “*Advanced Wavefront Engineering Via Polarization-Sensitive All-Dielectric Chiral Metasurface*” International Conference on Photonics and OSI, held on 23-25 October 2024, at CSIR-CSIO, Chandigarh, India.
80. Mukesh Kumar, **K. Nithyanandan**, “*Study on the optimization of laser spot size to improve the efficiency of beam combination systems*”, International Conference (Photonics, OSI Conference) held on 12–15 December 2024 ,IIT Kharagpur, Kharagpur.
81. Subrata Manna, Amala Jose,**K. Nithyanandan**, “*Experimental observation of complex lasing regimes in a mode-locked fiber ring laser*” International Conference on Photonics and OSI, held on 12 December 2024, IIT Kharagpur, Kharagpur, India.
82. Kushboo soni, **K.Nithyanandan**, “*Study on the effect of atmospheric turbulence on the phase synchronization in coherent beam combination through deep learning Optimizers*” at PHOTONICS 2024 held on 12–15 December 2024 ,IIT Kharagpur, Kharagpur.
83. Mamatha K L, **K. Nithyanandan**, “*Study of Propagation Dynamics of Optical Beam in an Inhomogeneous Nonlocal Nonlinear Medium*”, Conference on Nonlinear systems and Dynamics, held on 10–13 March 2025, Bharathidasan University, India.
84. Amala Jose, **K. Nithyanandan**, “*Optimization of Gain-flattened Filter for Ultra-broadband Tm-doped Fiber Amplifier*”, SPIE-Optics and Optoelectronics, held on 7–10 April 2025, Prague, Czech Republic.
85. Subrata Manna,**K. Nithyanandan**, “*Multiwavelength NPR-Based Mode-Locked Tunable and Switchable Fiber Laser: Design, Optimization, and Performance Analysis*” International

Conference on Photonics and OSI, held on 11 April 2025, Mahindra University, Hyderabad, India.

86. Khushboo Soni, **Nithyanandan Kanagaraj**, “*Adaptive Beam Steering in Coherent Beam Combining Systems Using Adagrad Optimization for Precision Applications*” IEEE Summer topical Meetings, held on 21–23 July 2025, Berlin, Germany.
87. Bharathy Jacob, **Nithyanandan Kanagaraj**, “*Polarization-Dependent Optical Switching and Complex Cylindrical Vector Beam Generation Using an All-Dielectric Chiral Metasurface*”, Advanced Photonics congress held on 13–17 July 2025, Marseille, France.
88. Pratiksha Meena, Sudharsan J B, **Nithyanandan Kanagaraj**, “Exploring the electronic, optoelectronic, and thermoelectric properties of novel half-Heusler TiXAs (X = Co, Rh): a DFT study” held on August 11-13, 2025, Sathyabama Institute of Science and Technology, Chennai, India.

## 5. List of Patents Published/Granted

❖ All published and granted patents are after joining IITH.

- Total No of Patents : **15**
- No. of Patents Granted : **2**
- No. of Patents Published : **9**
- No. of Patents Filed : **4**

| S.NO | Inventors                                                                                               | Title                                                                                                               | Patent Number | Status  |
|------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------|---------|
| 1    | <b>Nithyanandan Kanagaraj</b> ,<br>Bharathy Jacob,<br>Akash Domnic<br>Thomas,<br>G. Nagarajan           | A Method and a Wavelength Demultiplexing Diffractive Neural Network System to Reconstruct Unique Intensity Patterns | 569848        | Granted |
| 2    | Yashwanth Mohan<br>Kumar,<br>Harishankar<br>Manoharan, Vishnu<br>Unni,<br><b>Nithyanandan Kanagaraj</b> | Device and a methodology to determine and control the state of coherent sources Using high-speed photodetectors     | 202441040191  | Granted |

|   |                                                                                                            |                                                                                                        |              |           |
|---|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------|-----------|
| 3 | <b>Nithyanandan Kanagaraj</b> ,<br>Bharathy Jacob                                                          | A Method and System for Programmable Laser Beam Shaping using a Diffractive Neural Network             | 202541004549 | Published |
| 4 | <b>Nithyanandan Kanagaraj</b> and<br>Amala Jose                                                            | A method and a system of operation in a fiber amplifier for achieving a flattened gain                 | 202541061141 | Published |
| 5 | G. Nagarajan,<br>Swarubini PJ,<br>Kaveti Pavan<br>Kumar,<br>Karthick PA,<br><b>Nithyanandan Kanagaraj</b>  | AIKED: System and Method for AI-based Keratoconus EyeCondition Determinations                          | 202441036963 | Published |
| 6 | G. Nagarajan,<br><b>Nithyanandan Kanagaraj</b> ,<br>Shrikanth<br>Bharadwaj,<br>M. Divyadarshini            | System and Method for AI-Powered Non-Contact Sensing-based Stress Detection                            | 202441038983 | Published |
| 7 | G. Nagarajan,<br>Krushna Devkar,<br>Arun<br>Kumar, Shrikanth<br>Bharadwaj<br><b>Nithyanandan Kanagaraj</b> | System and method for Real-time Keratoconus Screening and assessment using Portable AI Handheld Device | 202441097171 | Published |
| 8 | <b>Nithyanandan Kanagaraj</b> ,<br>Khushboo Soni,<br>John Rozario<br>Jegaraj                               | Methods and systems for dynamic beam shaping and sequencing in a coherent beam combination system      | 202541087214 | Published |
| 9 | <b>Nithyanandan Kanagaraj</b> ,<br>M. Dinesh Kumar                                                         | Transformer-based Encoder-Decoder framework for modeling system dynamics                               | 202541070936 | Published |

|    |                                                                    |                                                                                                                             |              |                    |
|----|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|
| 10 | <b>Nithyanandan Kanagaraj</b> , M. Dinesh Kumar, and Khushboo Soni | Real-time phase locking in laser arrays                                                                                     | 202541103740 | Filed              |
| 11 | <b>Nithyanandan Kanagaraj</b> and Amala Jose                       | A method for flat-gain broadband application in a multi-pass optical amplifier configuration                                | 202541101608 | Filed              |
| 12 | <b>Nithyanandan Kanagaraj</b> and Subrata Manna                    | Engineering Tunable and Switchable Multiwavelength Emission in Mode-Locked Fiber Lasers via Nonlinear Polarization Rotation | -            | Filling-in Process |
| 13 | <b>Nithyanandan Kanagaraj</b> and Subrata Manna                    | Compact NPR-Mode-Locked All-Fiber Laser for Tunable and Reconfigurable Hybrid Soliton Generation                            | -            | Filling-in Process |
| 14 | <b>Nithyanandan Kanagaraj</b> and Dinesh Kumar                     | Autonomous Multi-Sensor Target Recognition and Tracking System for Directed Energy Systems (DES)                            | -            | Filling-in Process |
| 15 | <b>Nithyanandan Kanagaraj</b> and Amala Jose                       | Methods and processes for Broadband amplification in a few mode fiber amplifiers through gain-guiding mechanism             | -            | Filling-in Process |

## 6. List of Research Projects and Grants

Total Running Projects – **13**

No. of Projects Completed - **2**

Funding Secured to Date: ~ **13 Cr**

Funding pending Sanction - **11.5 Cr**

| S No | PI/ Co-PI | Project Title                                                                                                                                 | Sponsoring Agency                  | Project Period | Duration | Project Cost (in Lakhs) |
|------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------|----------|-------------------------|
| 1    | PI        | Optimization of fiber laser cavity using an evolutionary algorithm to investigate novel lasing states                                         | IITH                               | 2021-23        | 2        | 25                      |
| 2    |           | Design and Development of a multifunctional all-fiber laser system for the “On-demand” generation of structured laser light                   | JICA                               | 2022-25        | 3        | 20                      |
| 3    | PI        | Towards Multifunctional Two-micron Ultrafast Fiber Laser for High-Precision Biomedical applications (TUFL)                                    | CEFIPRA                            | 2023-26        | 3        | 180                     |
| 4    | PI        | Harnessing optical non linearity and spatiotemporal dynamics to carve the higher-dimensional light bullets in multimode optical fiber systems | DST-CRG                            | 2024-27        | 3        | 40                      |
| 5    | PI        | Tensor-based Machine Learning for Early Detection of Keratoconus Eye Conditions (TENSORKATE)                                                  | IIT KGP (National AI Call) AI4ICPS | 2023-24        | 2        | 42                      |

|    |       |                                                                                                                                       |           |         |     |     |
|----|-------|---------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|-----|-----|
|    |       |                                                                                                                                       |           |         |     |     |
| 6  | PI    | Development of Next Generation of spatio-Temporal MultimodeBroadband Amplifiers for Fiber Optic Communication (NextSTAR)              | UKIERI    | 2024-27 | 3   | 120 |
| 7  | Co-PI | A Proof-of-Concept study on laser additive manufacturing of single crystal super alloys through spatio-temporally tailored laser beam | ARDB      | 2023-25 | 2   | 22  |
| 8  | PI    | Development of Dynamic Beam Engineering Technology for Laser-Assisted Advanced ManufacturingSystems                                   | ER&IPR    | 2024-27 | 3   | 627 |
| 9  | PI    | A PoC study on the intelligent vision system for coherent beam combined Target-in-the-Loop                                            | DRDO-CARS | 2024-25 | 1.5 | 84  |
| 10 | Co-PI | Development of a platform that would aid a nano-ornithopter to learn to fly(LearnToFly)                                               | DFTM      | 2024-26 | 3   | 101 |
| 11 | PI    | Complexity and Nonlinear Dynamics in STEM                                                                                             | DST       | 2022-23 | 1   | 3.5 |
| 12 | PI    | Development of a compact HandheldAI-driven Intelligent IR imaging system for the Early Detection of Keratoconus                       | TDG-IITH  | 2024-26 | 2   | 37  |

|    |    |                                                                                                                                                                                      |             |         |   |                           |
|----|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|---|---------------------------|
| 13 | PI | Prototype Development of Generation of Spatio-Temporal Multimode Broadband Amplifiers                                                                                                | SPARC       | 2025-26 | 1 | 10                        |
|    | PI | IIT-DAAD Joint Exchange Fellowship                                                                                                                                                   | DAAD        | 2025-26 | - | Exchange Fellowship       |
| 14 | PI | Adaptive Optical Power Beaming through Tiled Fiber Array for Wireless Power Transfer (EMPOWER)                                                                                       | DFTM        | 2026-28 | 3 | 887<br>(To be Sanctioned) |
| 16 | PI | Development of a compact Intelligent All-optical photonic device based on Photonic Lantern for real-time feedback control aimed at wavefront sensing and Optical beam control (DIAL) | DFTM        | 2026-27 | 2 | 298<br>(To be Sanctioned) |
| 17 | PI | Development of algorithms for Autonomous Directed Energy Systems (DES)                                                                                                               | CHESS -CARS | 2026-28 | 2 | 148<br>(To be Sanctioned) |
| 18 | PI | Development of Coherent BeamCombining-based Dynamic Beam Engineering Technology for ext-Generation Laser-Assisted Manufacturing Co-BEAM                                              | DST-AMT     | -       | 2 | 324<br>(Submitted)        |

## 7. Research Guidance

Current UFO Group Strength - 21

Ph.D. Scholars: 13

Postdoctoral Fellows: 4

Project Fellows/Interns: 4



## 7.1 List of Ph.D Guidance

| S no | Name of the Student     | Area of Research                         | Funding         | Status of Ph.D.      |
|------|-------------------------|------------------------------------------|-----------------|----------------------|
| 1    | Mr. Rajkumar Joseph     | Smart Photonics                          | Industry Funded | OC in Month          |
| 2    | Miss.Amala Jose         | Ultrafast Fiber Laser                    | CSIR            | OC in Month          |
| 3    | Miss.Mamatha K L        | Multimode Photonics                      | CSIR            | Middle of the Thesis |
| 4    | Mr.Subrata Manna        | High-power fiber laser                   | ID - CIP        | Middle of the Thesis |
| 5    | Mr.Thirumugam S         | Physics Informed Neural Network          | Industry Funded | Middle of the Thesis |
| 6    | Miss.Pratisha Meena     | 2D Materials for Photonic application    | CSIR            | RPS Completed        |
| 7    | Miss.Bharathy Jacob     | Advanced Beam Engineering                | MoE             | RPS Completed        |
| 8    | Miss. Khushboo Soni     | Directed Energy Applications             | MoE             | RPS Completed        |
| 9    | Mr. Yashwant Kumar      | Autonomous Drones                        | ID-CIP          | Coursework           |
| 10   | Mr. S. Arun Kumar       | Edge Computing Healthcare devices        | Project Funded  | Coursework           |
| 11   | Miss. Pooja P M         | Structured lasers and amplifiers         | Project Funded  | Coursework           |
| 12   | Mr. Pravin Sharma       | Beam Director for Directed Energy Weapon | DRDO Sponsored  | Coursework           |
| 13   | Mr. Neeraj Ramachandran | Lasers for Advanced Manufacturing        | Project Funded  | Coursework           |

## 7.2 Details of the Postdoc Currently Working

| S NO | Name of the Student  | Area of Research                 | Funding   |
|------|----------------------|----------------------------------|-----------|
| 1    | Dr. Mukesh Kumar     | Incoherent Beam combining system | CEFIPRA   |
| 2    | Dr. Prasenjit Prahaj | Coherent Beam Combination        | ER&IPR    |
| 3    | Dr. Monish K         | Multimode Photonics              | CARS-DRDO |
| 4    | Dr. Krishna K. Singh | Optical Tweezers                 | CEFIPRA   |

## 7.3 Details of the Project Fellow and Interns

| S NO | Name of Project Fellow/Intern | Duration                   | Work / Project Area                                                                  |
|------|-------------------------------|----------------------------|--------------------------------------------------------------------------------------|
| 1    | Tanaya Karmakar               | Aug 2022 - April2023       | Supercontinuum Generation In Multimode Fibers                                        |
| 2    | Padmanava Chatterjee          | June 2023 - June 2024      | Diffraction Neural Network                                                           |
| 3    | Rishabh Singh                 | Jan-May 2023               | Beam Combining using SPGD                                                            |
| 4    | Abhishek Tiwary               | August 2023 - April 2024   | Temperature sensors using Fiber Bragg Grating                                        |
| 5    | Aleena Antony                 | 15 May 2025 - 14 July 2025 | Design and Simulation of an Active Phase Change Metasurface for Tunable Beam Shaping |
| 6    | Renuka                        | July-2025 onwards          | Designing of optical                                                                 |

|   |         |                      |                                                                             |
|---|---------|----------------------|-----------------------------------------------------------------------------|
|   |         |                      | phased array in coherent beam combining using ZEMAX                         |
| 7 | Sreerag | May-2025 onwards     | Working on beam engineering — focusing on design, analysis, and performance |
| 8 | Dinesh  | January-2025 onwards |                                                                             |

## 8. Invited Talk delivered in

1. Invited Lecture on **“Laser Beam Engineering using Diffractive Neural Network”** at the Optica Advanced Laser Congress held on Oct 19-23, 2025 at Prague, Czech Republic.
2. Delivered a talk on **“Ultrafast 2 micron laser for Biomedical applications”** at the Institute of Photonics and Electronics, on Oct 25, 2025 at the Czech Academy of Sciences, Prague, Czech Republic.
3. Delivered a talk on **“Coherent beam combination for advanced manufacturing”** at the Institute of Photonics and Electronics on Oct 25, 2025 at the Czech Academy of Sciences, Prague, Czech Republic.
4. Delivered a talk on **“Recent developments in Broadband amplifiers”** at the University of Southampton under the UKIERI-SPARC initiative, August 25-26, 2025.
5. Delivered a lecture on **“Ultrafast Dissipative Soliton Dynamics”** at the University of Aberdeen, August 23-24, 2025, fostering interdisciplinary discussions with faculty and researchers.
6. Delivered a talk titled **“Recent Development in Ultrafast Fiber laser research at IIT Hyderabad”** at Interdisciplinaire Carnot de Bourgogne, Universite de Bourgogne, on July 9, 2025.
7. Invited lecture titled **“A framework on broadband thulium doped optical amplifier”** at Interdisciplinaire Carnot de Bourgogne, Universite de Bourgogne, on July 9, 2025.

8. Delivered a talk titled “**Experimental Observation of Coupled Dark-Bright Soliton Co-existing Regimes in Mode-Locked Fiber Lasers**” at CLEO-Munich, held from June 23–27, 2025.
9. Delivered an Invited talk titled “**Laser Beam Engineering using Diffractive Neural network**” at the CLEO-Munich , held from June 23–27, 2025, on June 25, 2025.
10. Delivered an invited lecture at the Faculty Development program titled “**Bridging Communication, Defence and Space Applications**” on June 16, 2025 at IIT Kurnool.
11. Delivered a National Science Day Lecture on “**Photonics for the Future: Bridging Communication, Defence and Space Applications**” on February 28, 2025, at Stanley College of Engineering and Technology, Hyderabad.
12. Invited lecture “**Ultrafast Fiber Optics: A Versatile Toolbox for Photonic Intelligence**” at the International Conference on Photonics (Photonics 2024) at IIT Kharagpur on Dec 13, 2024.
13. Delivered a lecture titled “**Light Up Learning: Inspiring Young Minds Through Optics**” at the Navodaya Teachers Training Programme held at IIT Hyderabad on 25th Nov 2024.
14. Delivered an Invited Lecture titled “**Coherent Beam Engineering based Dynamic “Beam Engineering Advanced Manufacturing”**” at the Structured Light and Matter Interaction Workshop on Oct 25, at Osaka, Japan.
15. Delivered a Keynote lecture at the National Technology Day event at DRDO-CHESS titled “**Emerging trends in Laser-based technologies for Smart Solutions**” on July 15, 2024.
16. Presented a Talk at the Board of Governors meeting of IIT Hyderabad titled “**Ultrafast Photonic Technology**”.
17. Invited Talk “ **Recent Trends in Fiber Optics**” during the Faculty Development Program organized by Osmania University, held on June 18<sup>th</sup> 2022.
18. Invited Talk “**Odyssey of light in nonlinear optical fiber**” as a part of the National Science Day celebration by St. Xaviers College, Thumba, Thiruvananthapuram, on March 4, 2022.
19. Invited Talk on the National Laser Symposium titled “**Recent Advances in the Dissipative Soliton Dynamics**” held on Jan 19 at BARC, Mumbai.

20. A talk on the “**Introduction to Fiber Optics**” at the Government College, Meerut, on Dec 4, 2021
21. Invited Talk titled “**Perspectives in Fiber Optics**” on 16<sup>th</sup> May 2021 as a part of International Day of light celebration at the St Joseph University, Nagaland, India.

## 9. Book/Book Chapter Published

1. Published book entitled “**Modulational Instability and Supercontinuum Generation**” LAP LAMBERT Academic Publishing, ISBN **978-3-659-76383-0**, 248 pages, August 2015.
2. The Book chapter entitled “A theoretical study on Modulational Instability in relaxing Saturable nonlinear optical media” by K. Porsezian and **K. Nithyanandan** appeared in the book titled “**Odyssey of Light in Nonlinear Optical Fibers: Theory and Applications**” CRC Press ISBN **9781482236132** 568 Pages November 2014.
3. The Book chapter entitled “Extreme wave dynamics from incoherent dissipative solitons in fiber laser cavities” by Phillipe Grelu et al, “**Nonlinear Guided wave optics: A tested for extreme waves**” by IOP publishing Ltd. (In Press).

## 10. Awards, Achievements and Recognition

### (a) Distinguished National/International Award

1. **Awarded the Teaching Excellence Award** for the Year 2024 by IIT Hyderabad.
2. **Associate Editor** of the IEEE Photonics Technology Letters from 2025 onwards
3. **Conference Chair and Organizer** of Photonics 2026 at the IIT Hyderabad.
4. **Conference General Co-Chair** of the IEEE Photonics Summer Topical Meetings held on July 21-23, 2025 at Berlin, Germany.
5. **Founding General Secretary** of the Indian Society of Nonlinear and Complex Systems.
6. **Chair of the “Advances in the Laser-Assisted Manufacturing”** Panel Discussion at the Optica Advanced Laser Congress, held on Oct 21, 2025 at Prague, Czech Academy of Sciences.

7. **Chair of the “Light-by-Light Manipulation for next generation of Photonics system ”** the IEEE Photonics Summer Topical Meetings held on July 21-23, 2025 at Berlin, Germany.
8. **Chair of the “Ultrafast Dissipative Solitons”** at the IEEE Photonics Summer Topical Meetings held on July 22, 2025 at Berlin, Germany.
9. **Chair of the “Multimode Photonics”** IEEE Photonics Summer Topical Meetings held on July 21-23, 2025 at Berlin, Germany.
10. **Member of the Bureau of Indian Standard (BIS)** for the Fiber Optics and Application division.
11. Elevated to **Senior member** in the 2024 class of the SPIE – The International Society for Optics and Photonics.
12. Elected as the **Editor** of the Optical Society of India Newsletter.
13. Under my leadership, OPTICA Awarded our TG – Lasers in Manufacturing the **“Greatest Growth in Activity”** prize by the Board of Directors.
14. **Senior Member** of 2021 class of OPTICAL Society of America.
15. Become a part of the **3-Members Committee** of the Bureau of Indian Standards (BIS) in the Optics Section for the years 2021-23, 2024-26.
16. Vice-Chair of the OSA Technical Group **“Lasers in Manufacturing”** (Jan 2023 – Dec. 2025)
17. Events Officer of the OSA Technical Group **“Ultrafast Phenomena”** (Jan 2023 – Dec 2025)
18. Regional Coordinator of I-Stem by the Government of India from Dec 2021.
19. **Two Ph.D. Students** from my group were Awarded the prestigious **PMRF Fellowship**.
20. **Chair** of the OSA Technical Group in Laser Science in Manufacturing (Jan 2018 -Dec 2022).
21. Member of Bureau of Indian Standard-Optics Division from Oct 2021
22. Awarded **ROBERT S. HILBERT MEMORIAL TRAVEL GRANT** of the year 2014 in the amount of \$1,100 in recognition of the research excellence in the areas of Optical Engineering.
23. Awarded SPIE’s - **LASER TECHNOLOGY, ENGINEERING AND APPLICATION** Scholarship in the amount of \$5000 in the Year 2014.

24. Awarded **BUTI YOUNG SCIENTIST** of the Year 2015, Dec. 04, 2015 at Saha Institute of Nuclear Physics (SINP), Kolkata, India.
25. Awarded PBC Postdoctoral fellowship for outstanding Indian Students for the years 2019-2021 at Tel-Aviv University, Israel.
26. Awarded Department of Science and Technology (DST), “**National Post doctoral fellowship (NPDF)**” of the year 2016.
27. Awarded **CNRS Post-doctoral Fellowship** for the year 2016 at the Univ. of Burgundy, Dijon, France.
28. Qualified Council of Scientific and Industrial Research - National Eligibility Test (NET) June 2011.
29. Awarded **CSIR - Meritorious Senior Research Fellowship (SRF) from Government of India** – April 2013 – March 2016.
30. The article entitled “A Novel power play in the modulational instability induced Supercontinuum generation in the saturable nonlinear media” has been published in **KIRAN** (A bulletin of the Indian Laser Association) **Vol. No. 24, April 2013, Page No. 22.**
31. A research article “Modulational-instability-induced supercontinuum generation with saturable nonlinear response”, Phys. Rev. A, 82, 013825 (1-6) (2010) was selected for the October 2010 issue of **Virtual Journal of Ultrafast Science**, Vol.9, Issue 8/ Photonics.
32. The article entitled “Investigation of Supercontinuum generation via Modulational Instability in Exponential Saturable Nonlinear media” has been published in **KIRAN** – A bulletin of the Indian Laser Association **Vol 25, No.1, April 2014, Page No. 24.**
33. **Awarded Officer Travel grant from OSA** to attend the Student Leadership Conference and FIO/LS - 2013 at Orlando, Florida.
34. **Awarded International Student Travel Grant** to present a tutorial in the KOALA conference held on 24<sup>th</sup> – 29<sup>th</sup> November, 2013.
35. **Awarded Travel grant** with full support including, lodging and boarding to participate in the ICTP Winter College on Optics: Fundamentals of Photonics - Theory, Devices and Applications to be held on 3 – 21, February 2014, at Trieste, Italy.
36. **AWARDED** Euroscience International Travel Award in the amount of € 1300 to participate in the European Science Open Forum.

37. Awarded OSA **Foundation Student Travel Grant** in the amount of \$ 1500 for OSA Topical Meeting on Advanced Photonics held on 28 – 31 July 2014 at Barcelona, Spain.
38. **Awarded Travel Grant by Japan Society for Applied Physics (JSAP)** for participating in the 75<sup>th</sup> autumn meeting JSAP-OSA held on 17-21 September, 2014 at Hokkaido, Japan.
39. Awarded OSA **FOUNDATION STUDENT TRAVEL Grant** in the amount of \$ 1500 to participate in Frontiers in Optics (FIO -2014) held on 19 – 23 Oct. 2014 at Tucson, Arizona, USA.
40. Awarded **INTERNATIONAL STUDENT TRAVEL GRANT** to present a tutorial in the KOALA conference held at University of Adelaide, Adelaide on 23<sup>th</sup> – 28<sup>th</sup> November, 2014.
41. Awarded OSA **FOUNDATION STUDENT TRAVEL Grant** in the amount of \$ 1500 to participate in Frontiers in Optics (FIO -2015) to be held on 17 – 21 October 2015 at San Jose, California, USA.
42. Our article Featured Cover image in the Vol 49 of December 2016 issue of Journal of Physics B.
43. **Awarded Travel Grant by Japan Society for Applied Physics (JSAP)** for participating in the 77<sup>th</sup> autumn meeting JSAP-OSA held on 13-16 Sept, 2014 at, Japan.
44. Our article Featured Cover image in the Vol 93 of Feb. 2016 issue of Physical Review A.
45. **Nominated to represent INDIA**, as a part of the International Year of light celebration final report held at **UNESCO** head quarters at Paris, Oct 3, 2016.
46. Invited to the inauguration **of International Day of Light (IDL)** held on May 16 at UNESCO, Paris.
47. Our paper entitled “*Real-Time Observation of Internal Motion within Ultrafast Dissipative Optical Soliton Molecules*” published in Physical Review Letters has been selected for “**Paper of Editor’s suggestion**” and “**Featured in Physics**”. The article was covered in the “**News and Commentary**” with a special Editor written Synopsis. The article was also covered by Science Magazine.



## **(b) Best Paper/Oral/Poster Presentation Awards**

1. **AWARDED EXCELLENCE IN ORAL PRESENTATION** in “XXXVI OSI Symposium on Frontiers of Optics and Photonics” held on December 3-5 (2011) at IIT Delhi.
2. **AWARDED BEST CONTRIBUTORY PAPER** in “IONS” conference held on December 1-2 (2011) at IIT Delhi.
3. **AWARDED BEST POSTER PRESENTATION** in the “National conference on nonlinear dynamics and system (NCNSD)” held on July 12-15 (2012) at Indian Institute of Science Education and Research (IISER), Pune, India.
4. **AWARDED BEST POSTER PRESENTATION** in “37<sup>th</sup> Annual Symposium of Optical Society of India” held on January 23-25, 2013 at Pondicherry University, Puducherry.
5. **AWARDED BEST PAPER** in DAE-BRNS National Laser Symposium (NLS-21) held on February 6-9, 2013 at BARC, Mumbai, India.
6. **AWARDED 1<sup>st</sup> PRIZE** in the poster presentation in the International conference on “Perspectives on Nonlinear Dynamics” (PNLD - 2013) held on July 15 – 18, 2013 at HCU, Hyderabad, India.
7. **AWARDED BEST PRESENTATION** in the 8<sup>th</sup> International “Conference on Nonlinear System and Dynamics” (CNSD – 2013) held on December 11<sup>th</sup> – 14<sup>th</sup> 2013 at IIT Indore, India.
8. **AWARDED 4<sup>th</sup>** position in the best paper award during the Winter College on Optics: Fundamentals of Photonics - Theory, Devices and Applications to be held on 3 – 21, February 2014, at Trieste, Italy.
9. **AWARDED BEST POSTER PRESENTATION** in the International conference on Optics and Opto-electronics (ICOL -2014), held on March 05– 08, 2014 at Instrument Research and Development Establishment (IRDE), Dehradun, India.

## **(c) Books / Special Issues Edited**

1. **EDITOR** for the International Year of Light (IYL 2015) Newsletter in association with Optical Society of India (OSI), Indian Laser Association (ILA), Photonics Society of India (PSI), Luminescence society of India (LSI), Laser and Spectroscopy Society of India (LSSI), Indian Society for Radiation and Photochemical Sciences (ISRAPS).

2. **EDITOR** of the Proceedings of 1<sup>st</sup> Annual Symposium on “Recent Advances in Physics (RAP – 2014)” held on 9<sup>th</sup> April, 2014, at Dept of Physics, Pondicherry University, India .
3. **EDITORIAL ASSISTANT** of the proceedings of the – 27<sup>th</sup> PSSI National Symposium on Plasma Science & Technology on “Challenges of Power Generation & lighting 21<sup>st</sup> Century” held on December 10 – 13, 2012 at Department of Physics Pondicherry University, India.
4. **EDITOR** of the Proceedings of the International Conference on “**Complexity and Nonlinear Dynamics in STEM**” June 5-7, 2023 held at IIT Hyderabad.

## 11. Administrative Responsibilities

1. FiC, **Intellectual Property Facilitation Center (IPFC)** of IIT Hyderabad
2. Coordinated and successfully executed the **MoU between IITH-CGCRI**
3. **Founding Coordinator** of the M.Tech Ophthalmic Engineering Program
4. **Associated Faculty Member** of the School of Sustainability
5. **Founding Advisor of the Optical Society’s IIT H Student Chapter** (2021 - onwards).
6. **Founding Nodal Coordinator** for the **ISRO-START** program at IITH
7. **Convener of the International Conference** “Complexity and Nonlinear Dynamics in STEM”
8. **Mentor of Alfaleus**, a Start-Up company, by a student of the M.Tech Ophthalmic Engineering.
9. **Institute Laser Safety Officer** (From 2024)
10. **Coordinator of Rural Development Council** “School Teaching Activity”
11. **Coordinates of R&D Opportunities meetings** with Experts and Faculty members
12. **Co-Organizer of ICLA – 2023** organized by the Luminescence Society of India (LSI).
13. Regional Coordinator of the **I-Stem Hyderabad section**.
14. **Convener of the Campus Accommodation Committee** (Till-2024)
15. **Convener of the IITH Day Care Facilities** (Till Mid 2025)
16. **Director Nominee** for Special Drive Faculty Selection
17. Convener of Married Student Housing (Till 2024)
18. **Warden** of the Raman Hostel (Till Mid 2024)
19. **FiC of the Canteen Facility** of the IITH (Till 2024)
20. **Member** of the Mess committee (Till 2024)

21. **Member** and subject expert of the DAV Teacher recruitment
22. Member of the **Research Infrastructure Committee**.
23. **Faculty Advisor** of the 2021 M.Sc Physics
24. **Member of the Department Ph.D. admission committee** (2021-22)
25. Part of the department **asset verification committee** (2021 -22)
26. **Placement Coordinator** of M.Tech Ophthalmic Engineering
27. **Industrial Lecture Coordinator for Physics** (2022-2024) and Ophthalmic Engineering.
28. Coordinator of the **Department's monthly faculty meeting** (2022-2024).
29. **Coordinator of the Admission Committee** for M.Tech Ophthalmic Engg. (2023, 24, 25)
30. Represented ITH Cricket Team at the Inter-IITH Sports meet (2022, 23, 24).

## 12.Memberships/ Fellowship

1. Received a UKIERI Incoming Fellowship to visit the Optoelectronic Research Center (ORC), Southampton between March 5th to 15th, 2025.
2. Received an OPTICA grant to participate in the advanced Laser congress between Oct 20 -25, 2024, Osaka, Japan.
3. Got funding from JICA for the visit to Chiba University, Japan.
4. Early-career Professional Member of the Optical Society of America (OSA) (Membership Id – 1038854)
5. Member of the Optical Society of America Member Advisory Network.
6. Member of “The International Society for Optics and Photonics” (**SPIE**) (Membership Id – 3567267)
7. Life Member of “Optical Society of India” (**OSI**) (Membership Id – L 595)
8. Member of IEEE Photonics Society
9. Life Member of “Indian Physics Association” (**IPA**) (Membership No. GEN/LM/13003)
10. Life Member of “Indian Laser Association” (ILA) (Membership id – 1042).
11. Member of European Science Forum (EUOSCIENCE).
12. Life member of Plasma Science Society of India (PSSI).

### **13. Leadership – Conferences/Seminars/Workshops Organized**

1. **Conference Chair and Organizer** of Photonics 2026 at the IIT Hyderabad.
2. **Conference General Co-Chair** of the IEEE Photonics Summer Topical Meetings held on July 21-23, 2025 at Berlin, Germany.
3. **Chair of the “Advances in the Laser-Assisted Manufacturing”** Panel Discussion at the Optica Advanced Laser Congress, held on Oct 21, 2025 at Prague, Czech Academy of Sciences.
4. **Chair of the “Light-by-Light Manipulation for next generation of Photonics system ”** the IEEE Photonics Summer Topical Meetings held on July 21-23, 2025 at Berlin, Germany.
5. **Chair of the “Ultrafast Dissipative Solitons”** at the IEEE Photonics Summer Topical Meetings held on July 22, 2025 at Berlin, Germany.
6. **Chair of the "Multimode Photonics"** IEEE Photonics Summer Topical Meetings held on July 21–23, 2025, in Berlin, Germany.
7. **Chaired and organized the session** titled “Emerging Trends in Laser-Based Manufacturing: Panel Discussion” at CLEO-Munich, held from June 23–27, 2025.
8. **Organized the ISRO-START program** as Nodal Coordinator from April 3- May 21 2024 at IIT Hyderabad.
9. Organized **“I-Stem Awareness Workshop”** to promote the importance of equipment sharing.
10. Convener of Symposium on **“Nonlinear Optics and Photonics,”** June 7, IIT Hyderabad, India.
11. **Co-organized “Luminescence Science Society of India (LSSI) Conference** “Luminescence and its application (ICLA -2023), July 3-6, 2023 at IITH.
12. **Co-organizer of the ISRO, NAFEED-08<sup>th</sup> “National Finite Element Developers Meet”** 1<sup>st</sup> February, 2025, at IIT Hyderabad
13. Coordinated, in collaboration with DIA-IITH, the 75-Year Technology Roadmap 2047 for the Defence Sector—A Brainstorming Workshop on "Space Warfare and Space-Based Capabilities” on 22nd Sept. 2023.

14. **Chaired and Organized** the OSA Technical group – Lasers in Manufacturing Networking event “Recent Trends in Laser Technology and application in Manufacturing” Vienna 29 Sep – 03 Oct, 19.
15. Organizer of the “**The National Science Day**” at IIT Hyderabad on Feb 28, 2022.
16. **Chaired and Organized** the OSA Technical group – Lasers in Manufacturing Networking event “Recent Trends in Laser-based Manufacturing” Rochester 10<sup>th</sup> Oct, 2022.
17. Organizer of the “**The National Science Day**” at IIT Hyderabad on Feb 28, 2023.
18. Convener of International Conference on “Complexity and Nonlinear Dynamics in STEM” June 5-7, IIT Hyderabad, India.
19. **Founder Member** and First President of the Pondicherry University OSA Student Chapter.
20. **Coordinator** of International Year of Light (IYL 2015) events at Pondicherry University.
21. **Founder** of Pondicherry University Light Science Education (PULSE) Program.
22. **Founder** of Pondicherry University Physics Research Scholar’s Forum.
23. **Editor** of the compendium on the international year of light in India
24. In the team for the promotion of International Day of Light in India.
25. **Convener** of the First Annual Research Scholars meet on November 11, 2011 at Department of Physics, Pondicherry University, and Puducherry.
26. Member of the Organizing committee of “37<sup>th</sup> National Symposium of Optical Society of India (OSI)”, held on January 23<sup>rd</sup> – 25<sup>th</sup>, 2013 at Pondicherry University, Puducherry.
27. Organizing committee member of “**REFLECTION – 2013**” held on February 13 - 15, 2013, at Department of Physics, Pondicherry University, India.
28. Organizing committee member of “**INSPIRE SCIENCE CAMP – 2013**” held on April 5 -9, 2013, at Department of Physics, Pondicherry University, India.
29. Member of “**REFLECTION – 2014**” held on February 21 - 28, 2014, in, Pondicherry University.
30. **Convener** of “*Workshop on Fundamental Optics and Photonics (FOP)*” held on 31<sup>st</sup> January 2014 at School of Physical, Chemical and Applied Sciences, organized by Pondicherry University OSA Student Chapter, Puducherry, India.
31. **Convener** of 1<sup>st</sup> Annual symposium on “*Recent Advances on Physics*” (RAP 2014)” to be held on 9<sup>th</sup> April 2014, organized by Physics Research Forum in Collaboration with

**Pondicherry University OSA Student Chapter** at Department of Physics, Pondicherry University, and Puducherry, India.

32. **Organizing committee member** of 2<sup>nd</sup> Annual Symposium on Recent Advances in Physics (RAP 2015) held on 16-17 April, 2015 at Dept. of Physics, Pondicherry University.
33. Advisor and member of several student chapter events in India.

#### **14. Outreach Activities at IITH (if any)**

1. Organized Webinar titled “Recent Advances in Wire-Based Additive Manufacturing” by Yashwanth Kumar Bandari, on 09th Jan 2020 as part of OPTICA-Technical Group Lasers in Manufacturing.
2. Organized Webinar titled “Laser Aided Additive Manufacturing” by Sreekar karnati, on 20th April 2021 as part of OPTICA-Technical Group Lasers in Manufacturing.
3. Organized Webinar titled “Attosecond Pulses and Attosecond Spectroscopy by high-order Harmonic Generation in Gases” by Anne L'Huillier on 20th April 2021 as part of OPTICA-Technical Group Ultrafast Optical Phenomena.
4. Organized Webinar titled “Ultrafast transient absorption Spectroscopy by Giulio Cerullo on 12th Nov 2022 as part of OPTICA-Technical Ultrafast Optical Phenomena.
5. Organized Webinar titled “Directed Energy Deposition : What, How, and Why” by Melanie Lang, on 12th Nov 2022 as part of OPTICA-Technical Group Lasers in Manufacturing.
6. Organized Webinar titled “Next generation LiDAR for High-Priority Industrial metrology Applications” by Hollie Wright on 20th August 2023 as part of OPTICA-Technical Group Lasers in Manufacturing.
7. Organized Webinar titled “X-Photon 3D Lithography ” by Mangirdas Malinauskas on 16th Nov 2023 as part of OPTICA-Technical Group Lasers in Manufacturing.
8. Organized Webinar titled “Power Scaling in High Power Fiber Amplifiers” by Michalis Zervas, on 17th July 2023 as part of OPTICA-Technical Group Lasers in Manufacturing.
9. Organized a Webinar titled “Ultrafast Meets Ultrasmall: Lasers Writing at the Quantum Frontier” by Hong-Hua on 24th MAY 2024 as part of OPTICA-Technical Group Lasers in Manufacturing.
10. Organized Webinar titled “Attosecond Dynamics in molecules” by Mauro Nisoli, on 28th May 2024 as part of OPTICA-Technical Group Ultrafast Optical Phenomena.

11. Organized a Webinar titled “Ultrafast Dynamics of Excited States in Semiconductor Moire Superlattice” by Elaine Li on 09th July 2024 as part of OPTICA-Technical Group Lasers in Manufacturing.
12. Organized a Webinar titled “Hot Topics in Ultrafast Phenomena: Probing Optically Driven Quantum Materials” by Andrrea Cavalleri on 09th July 2024 as part of OPTICA-Technical Group Lasers in Manufacturing.
13. Organized a Webinar titled “Intelligent Light : Smart Lasers for Manufacturing” by Ben Mills on 17th June 2024 as part of OPTICA-Technical Group Lasers in Manufacturing.

### **15. List of Conferences/Schools/Workshops Attended**

1. Participated in Laser World of Photonics – India, Bangalore, 17–19 September 2025, engaging in industry–academia sessions on advances in photonics research and applications.
2. Attended the IEC General Meeting at India Habitat Centre, New Delhi, 8–9 September 2025, as a member of the Technical Committee on Fiber Optics Standards, contributing to discussions on multicore and hollow-core fiber technologies.
3. Participated in a research exchange visit under the UKIERI-SPARC initiative at the University of Southampton, 25-26 August 2025.
4. Completed the “**Refreshers Course in Experimental Physics**” from July 07-23, 2008 at Department of Physics, Pondicherry University, and Puducherry, India.
5. Attended the Winter School on “**Experimental Nonlinear Dynamics**” on December 1-5, 2008 at Saha Institute of Nuclear Physics, Kolkata, India.
6. Completed the “**DST-SERC school on Nonlinear Dynamics**” from January 4-26, 2011 at Bharathidasan University, Tiruchirapalli, India.
7. Attended the DST-SERC school on “**Guided Wave Optics and Devices**” from 7- 25, February 2011 at Central Glass & Ceramic Research Institute (CSIR-Lab), Kolkata, India.
8. Participated in the seminar on “**Concept of Theoretical Physics**” held on 23<sup>rd</sup> September 2005 as a part of the International Year of Physics celebration in the department of Physics, Pondicherry University, India.
9. Participated in the One day seminar on “**Einstein and his impact on Science**” held on 04<sup>th</sup> February 2006 at ion in the department of Physics, Pondicherry University, India.

10. Participated in the International Conference on **“Nano and Microelectronics”** (ICONAME) held on 5<sup>th</sup> Jan 2008 at Pondicherry Engineering college, Pondicherry, India
11. Attended the **“National Workshop on MATLAB and its applications”** in Science & Technology held on 9 – 10 October 2009 in VMKV Engineering college, Salem, India.
12. Participated in the **“Satellite Conference on Integrable Systems and Geometry”** held on August 12 -17, 2010, Department of Mathematics, Pondicherry University, Pondicherry, India.
13. Attended the **“National Workshop on Quantum Entanglement”** held on Feb. 17 – 19. 2010, Department of Physics, Pondicherry University, Puducherry, India.
14. Participated in the **“National TSPC Workshop on Nonlinear Physics: Theory, Experiments and Applications”**, held on March 29-31, 2010, at Nehru Memorial College, Puthanampatti, India.
15. Attended the **“National Seminar on Computational Material Science”** held on March 26, 2011, at Department of Physics, Pondicherry University, Puducherry, India.
16. Organized the Workshop on **“Fundamental of Optics and Photonics (FOP – 2014)”** held on Jan 31, 2014 at Dept. of Physics, Pondicherry University, and Puducherry, India.
17. Attended the ICTP Winter College on Optics: **“Fundamentals of Photonics - Theory, Devices and Applications”** to be held on 3 – 21, February 2014, at Trieste, Italy.
18. Attended the International workshop on **“Emerging Areas in Photonics and Future Applications (IWPPFA – 2015)”** held on 7 – 12, December, 2015 at Central Glass and Ceramic Research Institute, CSIR, Kolkata, India.
19. Completed the **“DST-SERC school on Lasers and Nonlinear Optics”** from March 22 – April 11, 2016 at Department of Physics, Pondicherry University, and Puducherry, India.

## 16.Colloborators

- (1) Prof. Phillippe Grelu, Universite de Bourgogne, Dijon, France.
- (2) Prof. Massmiliano Guasoni, Professor, University of Southampton, UK.
- (3) Prof. Kalaga Madhav, Leibniz Institute of Astrophotnics, Potsdam, Germany.
- (4) Prof. Takasagie Omatsu, Chiba University, Chiba, Japan
- (5) Prof. Senthil Ganapathy, Professor, University of Southampton, UK.
- (6) Prof. Patrice Tchofo Dinda, Universite de Bourgogne, Dijon, France.
- (7) Prof. Jose-Soto-Crespo, Instituto De Optica, Csic, Madrid.



- (8) Dr. Katrzyna Krupa, Università di Brescia, Brescia, Italy.
- (9) Dr. Hugues de Gaultet, Universite Joseph Fourier, Grenoble, France.
- (10) Dr. P. Muruganandam, Bharathidasan University, Tamil Nadu, India.
- (11) Dr. Zhiqiang Wang, Shanghai Jiao Tong University, Shanghai, China.
- (12) Dr. Aurelien Collet, Universite de Bourgogne, Dijon, France.
- (13) Dr. Vicente Duran Bosh Universitat Jaume I, Castellón, Spain.
- (14) Prof. Pavel Peterka, Institute of Photonics and Electronics, Prague.
- (15) Dr. Kyriacos Kalli, Cyprus University of Technology, Cyprus

## **17. Reviewer for Journals**

- (1) Nature Journals – Communications in Physics, Light Science & Applications
- (2) OSA - Optics Express, Optics letters, Josa B
- (3) SPIE - Optical Engineering, Journal of Nanophotonics,
- (4) IEEE - Photonics Technology letters, Photonics Review
- (5) Elsevier – Optical Communications, Optical fiber Technology, Optik, Journal of Advanced Research
- (6) Taylor & Francis – Journal of Modern Optics
- (7) Springer – Optics and Quantum Electronics
- (8) IOP – Applied Physics research, Journal of Physics commun., Journal of Phys. Photonics

## 8. Countries Visited Abroad

| s.No | Name and Place of the Country                                        | Purpose of the Visit                                                                                                               | Duration of the visit |
|------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1    | University of North Carolina, Charlotte, United States of America    | Presented paper in the International OSA Network of Students (IONS).                                                               | Oct. 2 – 4, 2013      |
| 2    | Hilton, Orlando, USA                                                 | Participation in the 97 <sup>th</sup> Optical Society of America (OSA) Annual Meeting Frontiers in Optics (FIO)/ Laser Application | Oct. 6 -10, 2013      |
| 3    | University of Sydney, Sydney, Australia                              | Presented Paper in the Conference on Optics Atoms and Laser Applications (KOALA).                                                  | Nov.24 – 29, 2013.    |
| 4    | International Center for Theoretical Physics (ICTP), Trieste, Italy. | Participation in the Winter College in Optics: Fundamentals of Photonics– Theory, Devices and Application                          | Feb. 2 – 22, 2014     |
| 5    | Carlsberg, Copenhagen, Denmark                                       | Participation in European Science Open Forum                                                                                       | June 21 – 26, 2014    |
| 6    | Barcelona, Spain                                                     | Participation in the Topical Meeting on Advanced Photonics                                                                         | July 27 – 31, 2014    |
| 7    | Santiago de Compostela, Spain                                        | Participation and Presented paper in the 23 <sup>rd</sup> Congress of International Commission for Optics (ICO - 23).              | August 26 -29, 2014   |
| 8    | Tucson, Arizona, United States of America                            | Participation and Presented paper in the 98 <sup>th</sup> Frontiers in Optics (FIO 2014) .                                         | Oct. 19 – 23, 2014    |
| 9    | University of Adelaide, Adelaide, Australia                          | Presented Paper in the Conference on Optics Atoms and Laser Applications (KOALA).                                                  | Nov. 23-28, 2014      |
| 10   | San Jose, California, USA                                            | Participation and Presented paper in the 99 <sup>th</sup> Frontiers in                                                             | Oct. 18-22, 2015      |

|    |                                                           |                                                                                                         |                     |
|----|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------|
|    |                                                           | Optics/Laser Science (FIO 2015) .                                                                       |                     |
| 11 | UNESCO Head Quarters at Paris, France.                    | Nominated to represent India as a part of the IYL concluding event                                      | Oct- 3, 2016        |
| 12 | ICB, Uiversite de Bourgogne, Dijon, France.               | CNRS Post-Doctoral Fellow                                                                               | July 2016 -Oct 2017 |
| 13 | UNESCO Head Quarters at Paris, France.                    | Invited to participate in the Inauguration of International day of light (IDL)                          | May- 16, 2018       |
| 14 | Washington DC, USA                                        | Participation and Presented paper in the 102 <sup>nd</sup> Frontiers in Optics/Laser Science (FIO 2015) | Sept 16 -20, 2018   |
| 15 | LIPHy, Uiversite de Grenoble-Alpes, Grenoble, France.     | CNRS Post-Doctoral Fellow                                                                               | Nov'17-Nov' 18      |
| 16 | Prague, Czech Republic                                    | Participation and Presented paper in the SPIE Optics and Optoelectronics                                | Apr'1-4, 2019       |
| 17 | Munich, Germany                                           | Participation and Presented paper in the CLEO Europe 2019                                               | June 23 – 27, 2019  |
| 18 | Institute of Photonics and Electronics                    | Post-Doctoral Fellow                                                                                    | Jan – Aug, 2019     |
| 19 | Optoelectronic Research Centre, Southampton               | Research Fellow                                                                                         | Aug19 – Dec 20      |
| 20 | Rochester, USA                                            | Event Chair of the Technical Group event at the Frontiers in Optics/Laser Science (FIO 2023)            | Oct 09 – 23, 2022   |
| 21 | University of Southampton, Optoelectronic Research Center | Organized a workshop on “Multimode Optics : From Optical fibers to Nanophotonics:                       | March 08-19,2025    |
| 22 | Visit to Power photonics, Edinburgh, Scotland             | To discuss on the Optical Phase Array and co development of CBC based manufacturing system              | March 18 ,2025      |

|    |                                                                  |                                                                                                                             |                       |
|----|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 23 | Dijon, France                                                    | Bilateral Collaborative Research Visit to ICB, Dijon through IFCPAR/CEFIPRA Project                                         | June 19- July 26 2025 |
| 24 | Munich, Germany                                                  | Delivered an Invited Talk<br>Chaired a Panel discussion on recent trends in advanced manufacturing<br>Presented a Oral Talk | June, 23 - 27, 2025   |
| 25 | Marseille, France                                                | Presenting paper and Organizing a networking event                                                                          | July 13 -17, 2025     |
| 26 | Berlin, Germany                                                  | Co-Chair, IEEE summer sumit on Photonics.<br>Delivering an Invited Talk<br>Session Chair                                    | July 21-23, 2025      |
| 27 | Leibniz Institute of Astrophotonics, Potsdam, Germany            | Academic and collaborative visit                                                                                            | July 22,2025          |
| 28 | University of Aberdeen                                           | Scientific Collaboration/discussion and delivered a lecture                                                                 | Aug 22-24, 2025       |
| 29 | Edinburgh, UK                                                    | Industry Visit to PowerPhotonics                                                                                            | Aug 24, 2025          |
| 30 | University of southampton                                        | Research Exchange Visit UKIERI-SPARC<br>Scientific interaction and delivered a talk                                         | Aug 25-26, 2025       |
| 31 | Prague, Czech Republic                                           | Chair of the session<br>Delivered an Invited Lecture                                                                        | Oct 17-25, 2025       |
| 32 | Institute of Photonics and Electronics,Czech Academy of Sciences | Collaborative Visit<br>Discussion on MoU with IITH                                                                          | Oct 24-25, 2025       |

**(a) References**

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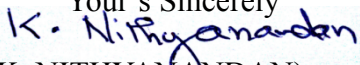
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I hereby declare that the above written particulars are true to the best of my knowledge and belief.

Date : 30-10-2025

Place: Telangana, India

Your's Sincerely  
  
(K. NITHYANANDAN)